Bitcoin Addresses

P2PKH or Legacy Address Format

- A P2PKH or legacy address, always starts with 1. For example, 1BvBMSEYstWetqTFn5Au4m4GFg7xJaNVN2.
- P2PKH stands for Pay-to-Pubkey Hash, which means paying to a hash of the recipient's public key.
- Legacy addresses are not SegWit compatible, but Bitcoin from a P2PKH address can be sent to a SegWit address without any problems.
- Transactions from a P2PKH address typically have higher fees due to their larger size.

Bitcoin Addresses

P2SH Address Format

- P2SH addresses have a structure similar to P2PKH addresses but start with a 3, for example 3J98t1WpEZ73CNmQviecrnyiWrnqRhWNLy.
- P2SH stands for Pay-to-Script Hash and enables more complex functionality than legacy addresses. It is commonly used for multisig addresses, where multiple digital signatures are required to authorize a transaction.
- P2SH addresses are also used for nonnative SegWit transactions through P2WPKH-in-P2SH. For most users, the important thing is that this address type is widely supported and can be used to send funds to both P2PKH and bech32 addresses.

Bitcoin Addresses

Bech32 Address Format

- Bech32 addresses have a distinct appearance, starting with "bc1," and are longer than legacy or P2SH addresses due to the prefix.
- Bech32 is the native SegWit address format, supported by most software and hardware wallets, but not fully embraced by all exchanges.
- While less than one percent of Bitcoin is currently stored in bech32 addresses, this number is gradually increasing.

Bitcoin Addresses

Bitcoin Cash Address Formats

Bitcoin Cash (BCH) addresses can follow either the legacy format (starting with a 1) or the Cash Address (Cash Addr) format.

The Cash Addr format is based on bech32 and starts with "q" or "bitcoincash:q."

BCH wallets can support both formats, allowing users to switch between them.

The primary reason for using the Cash Addr format is to distinguish BCH from BTC (Bitcoin) and prevent funds from being sent to the wrong address.

Wallet

A Bitcoin wallet is a software application or hardware device that is used to store, send, and receive Bitcoin. It is essentially a digital wallet that holds a user's private keys, which are used to access their Bitcoin holdings on the blockchain.

There are several types of Bitcoin wallets, including: –

Software wallets: These are digital wallets that can be downloaded and installed on a computer or mobile device. They can be further categorized into desktop wallets, mobile wallets, and web wallets.

Hardware wallets: These are physical devices that store a user's private keys offline. They are considered to be the most secure type of Bitcoin wallet.

Paper wallets: These are physical copies of a user's private keys that are printed out on paper. They are also considered to be a relatively secure option, but they can be vulnerable to physical damage or theft.

Custodial Wallet

- A custodial wallet is a type of digital wallet where a third-party service provider manages and secures the private keys of the wallet on behalf of the user.
- Private keys are a crucial component of a cryptocurrency wallet as they enable users to access and manage their cryptocurrency funds. The service provider takes on the responsibility of securing the private keys and ensuring the safekeeping of user funds. This can be advantageous for users who are new to the world of cryptocurrency and may not be familiar with the technical aspects of managing their own private keys.
- However, custodial wallets also have some drawbacks. First, users must trust the service provider to keep their funds safe, which can be a risk if the provider is hacked or goes out of business. Second, users do not have full control over their funds as they must rely on the service provider to manage their private keys.

Non-Custodial Wallet

- A noncustodial wallet, also known as a self-custody wallet, is a type of digital wallet that allows users to have full control over their private keys and manage their cryptocurrency funds directly without the involvement of a third-party service provider.
- Unlike custodial wallets, where a service provider manages the private keys and stores users' funds on their behalf, noncustodial wallets store private keys locally on users' devices, such as a computer or smartphone. This means that the user is responsible for safeguarding their private keys and ensuring the security of their funds.

Software Wallet

- A software wallet is a type of digital wallet that is installed on a computer or mobile device and is used to store, manage, and transact with cryptocurrencies.
- It is a software application that interacts with a blockchain network to enable users to send and receive cryptocurrency payments.
- Software wallets come in two main types: desktop wallets and mobile wallets. Desktop wallets are software applications that are installed on a personal computer, while mobile wallets are applications that are installed on a smartphone or tablet.
- Software wallets offer users the flexibility to manage their cryptocurrency assets from their own devices, without the need for a third-party service provider.
- They also offer a high degree of security, provided that users take appropriate measures to protect their private keys and secure their device.

Digital Wallet Vs. Bank Account

Digital Wallets	Bank Accounts
Safer for online payments as scammers cannot access all funds	Prone to scams and potential loss of all funds
Easier and free to open without interaction with a bank	Requires bank interaction and may involve fees
Convenient to use with simple password and mobile phone authentication	Lengthy card details input required
Instantaneous transactions regardless of time or holidays	Transaction speed may vary and may be affected by bank processing times
Allows global payments and transfers regardless of location	Tied to a specific country or region
Can offer offline payment options with issued cards	Primarily used for online and in-person transactions

Bitcoin Transaction

- Bitcoin's primary function is to facilitate transactions.
- Bitcoin's other features are all built around making it possible to produce, broadcast, validate, and, finally, add transactions to the global database of transactions (the blockchain).
- In the Bitcoin network, monetary transfers are encoded in data structures called transactions.
- Bitcoin's blockchain, a decentralized global database that acts as a global double-entry accounting system, records all transactions publicly.

Life Cycle of Transaction

There are multiple steps involved in the creation of, validation of, and addition to the blockchain that constitute the lifespan of a Bitcoin transaction.

Creation and Broadcasting

The process of transferring money begins when an authorized document is created and signed by one or more parties. Once the transaction has been validated, it is broadcast on the Bitcoin network, where each node is responsible for validating and spreading the news of the transaction to other nodes.

Verification and Inclusion

A mining node checks the transaction and verifies its legitimacy. When a transaction has been validated, it is added to the blockchain by being included in a block of previous transactions. To be permanently recorded and recognized as legitimate by all participants, subsequent blocks (confirmations) must be used. A new transaction life cycle can begin when the money are spent by the new owner.

Creation of Transaction

The process of creating a Bitcoin transaction is similar to writing a paper check in several ways. However, there are some fundamental differences due to the decentralized and digital nature of Bitcoin. Let's break down the key points of the statement with suitable examples:

A Bitcoin Transaction Is Like a Check in Concept

Just like writing a check indicates the intent to transfer money but does not immediately move funds, a Bitcoin transaction is an unsigned document stating the sender's intention to transfer a certain amount of Bitcoin to a recipient.

> **Example:** Suppose Alice wants to send 1 Bitcoin (BTC) to Bob. She creates a transaction specifying that 1 BTC should be sent to Bob's Bitcoin address. However, until it is signed and broadcast to the network, this transaction is just a draft—similar to a written check that hasn't been signed or deposited.

Creation of Transaction

The Creator and Signer Can Be Different

In traditional banking, the person who writes a check may or may not be the one who signs it. Similarly, in Bitcoin, anyone can draft a transaction, but only the owner of the Bitcoin (i.e., the person with the private key) can sign it to make it valid.

> **Example:** A company's finance team may draft a Bitcoin transaction to pay a supplier. However, the company's CEO (who holds the private key) must sign the transaction to approve the payment. This is similar to how an accounts payable clerk can prepare a check, but only an authorized person can sign it.

Creation of Transaction

Bitcoin Transactions Do Not Specify a Bank Account

Unlike checks, which clearly state the source account, Bitcoin transactions do not have an explicit "account" from which funds are deducted. Instead, Bitcoin uses **UTXOs (Unspent Transaction Outputs)**—unspent amounts from previous transactions—to fund new transactions.

> **Example:** If Alice has received BTC from multiple sources in the past, her wallet will automatically select the necessary UTXOs to cover the amount she wants to send. There is no single "bank account" that holds all her BTC; rather, it is a collection of unspent outputs.

Creation of Transaction

Signing the Transaction Makes It Valid

A Bitcoin transaction must be cryptographically signed using the sender's **private key** before it is considered valid. This ensures authenticity and prevents fraud.

> **Example:** If Alice has created a transaction to send 1 BTC to Bob, she must sign it using her private key. If she does not sign it, the Bitcoin network will reject it as invalid.

Broadcasting to the Network and Adding to the Blockchain

Once a transaction is signed, it is broadcast to the Bitcoin network. Miners then verify and include it in the **blockchain**, the public ledger of Bitcoin transactions.

> **Example:** After Alice signs the transaction and broadcasts it, miners validate it and add it to a new block. Once the block is confirmed, Bob officially receives the 1 BTC.

Creation of Transaction

Bitcoin Transaction vs. Check Transaction

Feature	Paper Check	Bitcoin Transaction
Creation	A check is written with details (amount, payee, etc.).	A Bitcoin transaction is drafted with details.
Signing	The account holder signs the check to validate it.	The sender signs the transaction using their private key.
Source of Funds	Clearly mentions a specific bank account.	Uses UTXOs (unspent previous transactions) as inputs.
Verification	Bank processes and verifies before approving.	Bitcoin miners verify and include it in the blockchain.
Execution	The recipient deposits it; the bank transfers funds.	The network confirms the transaction, transferring BTC.

Broadcasting Bitcoin Transaction

- Delivering a Bitcoin transaction to the network guarantees that it will be seen by many nodes and added to the blockchain.
- Thousands of Bitcoin nodes need to receive a small data structure of about 300–400 bytes, which represents a transaction.
- As long as many nodes are employed to guarantee propagation, senders are not required to trust the nodes used for broadcasting.
- Since the transaction is signed and does not include any private information, nodes do not need to trust the identity of the sender.
- Unlike credit card transactions, which require encrypted networks, Bitcoin transactions can be broadcast publicly via any suitable network transport technology.

Propagating Transactions on the Bitcoin Network

- When a Bitcoin transaction is delivered to a node on the Bitcoin network, that node is responsible for verifying the legitimacy of the transaction.
- If the transaction is genuine, the node will forward it to other nodes in the network and signal back to the sender that the transaction was successful.
- In the event of an invalid transaction, the node will send a rejection message back to the sender.
- Through a technique known as flooding, transactions are broadcast from a central node to all of the peer nodes with which it communicates. Once a transaction is processed and accepted by the network, it spreads out to every node in an exponential fashion. The Bitcoin network is made to spread transactions and blocks quickly and reliably, protecting against spam, denial-of-service attacks, and other possible problems. Each node checks transactions on its own, which stops bad transactions from growing past a single node.